

2 Operators and Control Structures

2.1 Assignment operators

The following code assigns the value 15 to the variable x.

```
int x;  
x = 15;
```

In Java, a single equals sign is the assignment operator. Consider we append the statement with:

```
x = x + 5;
```

The new value of x will now be 20. First, the addition on the right is done. Then, the result is assigned to the the variable on the left. In Java, you can also use the following shortcut to arrive at the same value:

```
int x = 15;  
x = x + 5;
```

2.2 Increment/Decrement operators

There is one addition statement in Java that is so common, it gets its own operator. It means, simply, “Add/Subtract one to the variable.” And you can do pre-increment/decrement, or a post-increment/decrement.

```
Pre  
++x; x++;  
Post  
--x; x--;
```

If each of these statements is on a line by itself, there is no difference in the effect of doing a pre- or post-increment. However, if the variable that is having the pre- or post-increment applied to it is used within another statement, your choice of pre- or post-increment can alter your results which will be made clear using this example.

```
// PrePostIncrement.java  
public class PrePostIncrement  
{  
    public static void main( String args[] )  
    {  
        int x=0;  
        System.out.println( " Baseline, x = " + x );  
        System.out.println( "\n Pre-increment = ++x = " + ++x );  
        System.out.println( "\n After increment, x = " + x );  
    }  
}
```